Desktop Hardening with Group Policy

Cybersecurity Portfolio Project

1. Project Overview

This project focused on improving Windows desktop security using Local Group Policy settings. The project demonstrates how administrators can apply centralized security controls to strengthen endpoint protection.

2. Objective

The goal of this project was to harden a Windows system by reviewing and configuring Group Policy security settings related to user restrictions, desktop controls, and administrative templates.

3. Tools Used

- Windows 10
- Local Group Policy Editor (gpedit.msc)
- VirtualBox

4. Skills Demonstrated

- Windows Hardening
- Endpoint Security
- Group Policy Management
- Security Baselines
- Access Restrictions
- Administrative Security Controls

5. Lab Environment

The lab environment was created inside a virtual machine using VirtualBox. A Windows 10 system was used to configure and review Local Group Policy settings.

6. Steps Performed

1. Opened the Run dialog box.
2. Launched Local Group Policy Editor using gpedit.msc.
3. Navigated through Administrative Templates.
4. Reviewed desktop and user restriction settings.
5. Examined security-related configuration options.
6. Documented the hardening process.

7. Screenshots

Figure 1 — Opening Group Policy Editor

- Opened the Run window.
- Entered gpedit.msc.
- Launched Local Group Policy Editor.

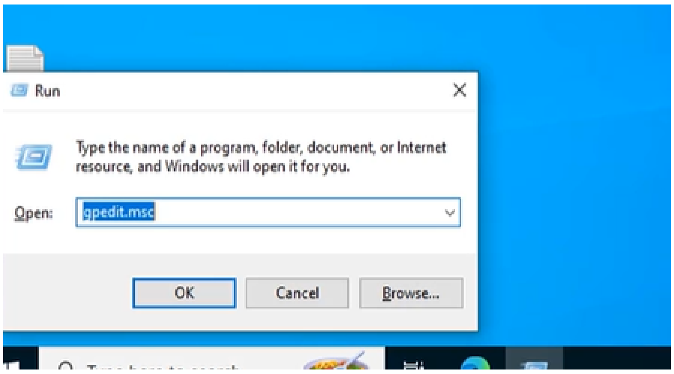
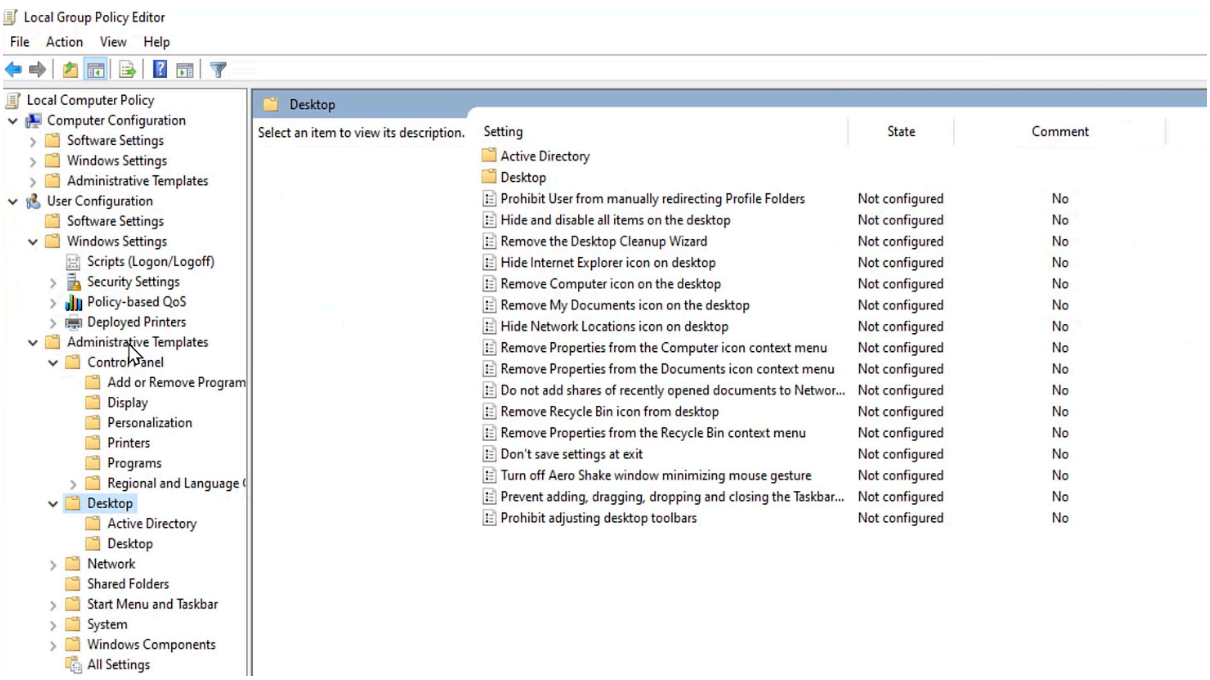


Figure 2 — Reviewing Administrative Template Settings

- Navigated through:
  - User Configuration
  - Administrative Templates
  - Desktop settings
- Reviewed multiple hardening and restriction policies.



8. Security Benefits

Using Group Policy improves endpoint security by enforcing security standards, restricting risky behavior, and strengthening system configurations.

Security hardening helps:

- Reduce unauthorized changes
- Improve compliance
- Strengthen desktop security
- Enforce administrative controls

9. What I Learned

I learned how Windows Group Policy can be used to apply enterprise security controls and improve endpoint hardening practices.

I also gained experience navigating administrative templates and reviewing desktop restriction policies used in enterprise environments.

10. Conclusion

This project demonstrated practical cybersecurity skills related to Windows security hardening, administrative policy management, and endpoint protection using Local Group Policy Editor.